SafeCharge-RL

Reinforcement learning for EV charging when drivers' departure times are wrong or misreported

Papers cheat sheet: what each referenced paper does, what we take from it, and where it falls short. Facts are from our proposal (Sections III-VI); read the abstracts before the review.

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Two papers matter most. Chen and Shi (2022) is the closest to our method (Lagrangian constrained RL). **Ferragut et al. (2021)** is the closest to our question (drivers who under-report get more service, and the fix of serving only until the declared departure).

The seven papers

Paper	What it does	What we take	Where it falls short
1. Li, Wan, He (2020), IEEE Trans. Smart Grid 11(3), 2427-2439	Single-EV charging as a CMDP; safe deep RL keeps charging within limits at low cost with uncertain prices and departure	The CMDP framing	One charger, no shared grid cap; safety is learned, not guaranteed
2. Chen, Shi (2022), arXiv 2209.09772	CMDP solved with augmented Lagrangian + Soft Actor-Critic, double critic; compares cost with a perfect-information MPC	Lagrangian budgets; benchmark against a perfect-information optimum	One EV, hourly steps, synthetic arrivals and departures; no hard-constraint layer
3. Ting et al. (2024), arXiv 2412.18047	Hierarchical multi-agent actor-critic for workplace charging with a station limit, dynamic prices, early departures; critic augmentation for departure uncertainty; compared with an LP oracle	Departure uncertainty matters; oracle comparison	Simulated EVs; capacity handled by a penalty; no comparison with EDF/LLF rules
4. Li, Sun (2023), arXiv 2311.05941	OOD-aware charging on real Caltech data: behaviour shifted after a pricing change and COVID; blends DDPG with MPC; shows user-entered departures and capacities differ from true values	Evidence that reports are unreliable and that RL degrades after a shift	Two chargers, quadratic cost, DDPG; not about misreporting
5. Ferragut, Narbondo, Paganini (2021), IFIP Performance	EDF and LLF with uncertain departures (mean-field model): under-reporting vehicles get more service; fix: serve only up to the declared departure; real traces	Hypotheses H1 and H3; the curtail fix	Rule-based only, no learning, no prices
6. Gabriele et al. (2026), ACM e-Energy, arXiv 2606.19199	Departure time not available to RL; forecaster trained end-to-end with the agent (decision-focused RL); up to 14% more reward, 55% less unsupplied energy	Recent example of departure uncertainty in RL; a forecast could replace the report later	Short paper; departure missing, not wrong or strategic
7. Lee, Li, Low (2019), ACM e-Energy, pp. 139-149	Introduces and analyses ACN-Data (Caltech and JPL sessions)	Our dataset	Not an RL paper

Products and tools mentioned

- **ChargePoint power management:** configured limits, schedules, sharing rules; not a published learned policy.

- **Enel X JuiceNet:** grid-signal-based management; algorithms unpublished; Enel X Way closed its North American business in Oct 2024.
- **Caltech ACN:** real multi-port system using MPC with driver-entered departures; source of ACN-Data and ACN-Sim.
- **OCPP:** message standard for sending charging profiles; does not say how limits are computed.
- **ACN-Sim and EV2Gym:** open-source simulators. We built our own small numpy environment instead (ACN-Sim works in amps and its data client failed authentication).

What the literature says together (the 30-second version)

- Safe and constrained RL for EV charging is established, so a safety layer alone is not new.
- Comparisons with a full-information optimum exist, but on synthetic or simulated cars.
- Uncertain departures are treated as noise or missing data; Li and Sun show real reports are unreliable; Ferragut shows under-reporting pays under rules.
- **Open question we study:** can a driver game a learned policy, on real sessions, with a shared hard grid cap?

Likely questions about the papers

Question	Answer
Which paper is closest to yours?	Chen and Shi for the method (Lagrangian CMDP); Ferragut et al. for the question (misreporting and the curtail fix).
Why not compare with Li and Sun's OOD-Charging?	It uses two chargers, DDPG and a quadratic cost, and focuses on distribution shift, not deadlines or misreporting. Not comparable on our setup.
What do you take from Ferragut?	H1 (under-reporting gains) tested on a learned policy, and the curtail fix for H3. Their result carried over: with curtailment every method loses by under-reporting.
Is the safety projection new?	No. We say that explicitly. The new part is the evaluation: a learned policy under misreporting on real multi-EV data.
Why PPO-Lagrangian and not SAC-Lagrangian like Chen and Shi?	PPO is stable and simple for our two costs; SAC-Lagrangian is planned only if time allows.
What background do I need?	CMDP (MDP with cost budgets), Lagrange multipliers and dual ascent, PPO, EDF/LLF/MPC, and what a hindsight-optimal LP is.